FRONT AXLE - 181FBI (Continued)

(4) Place Arbor Disc 6732 on Arbor D-115-3 in position in axle housing side bearing cradles (Fig. 12). Install differential bearing caps on arbor discs and tighten cap bolts to 41 N·m (30 ft. lbs.).

**NOTE: Arbor Discs 6732 has different step diameters to fit other axles. Choose proper step for axle being serviced.**

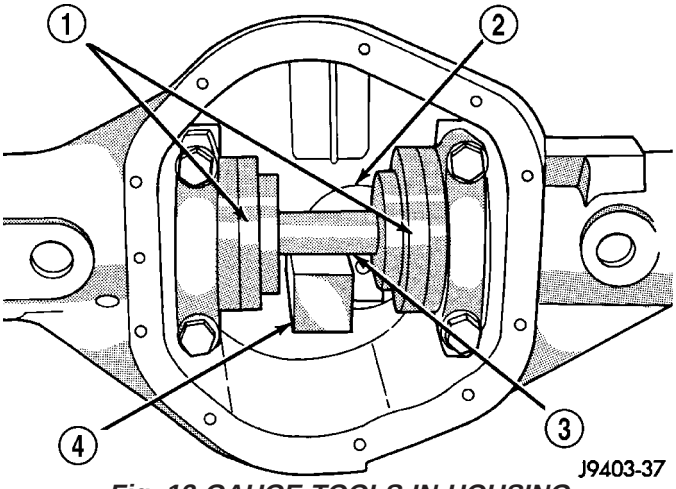



*Fig. 12 GAUGE TOOLS IN HOUSING*

1 - ARBOR DISC
2 - PINION BLOCK
3 - ARBOR
4 - PINION HEIGHT BLOCK

(5) Assemble Dial Indicator C-3339 into Scooter Block D-115-2 and secure set screw.

(6) Position Scooter Block/Dial Indicator flush on the pinion height block. Hold scooter block and zero the dial indicator.

(7) Slowly slide the scooter block across the pinion height block over to the arbor (Fig. 13). Move the scooter block till the dial indicator probe crests the arbor and record the highest reading.

(8) Select a shim/oil baffle equal to the dial indicator reading plus the pinion depth variance number etched in the face of the pinion (Fig. 8). For example, if the depth variance is –2, add +0.002 in. to the dial indicator reading.

## DIFFERENTIAL SIDE BEARING PRELOAD AND GEAR BACKLASH

Differential side bearing preload and gear backlash is achieved by selective shims positioned behind the differential side bearing cones. The proper shim thickness can be determined using slip-fit Dummy Bearings D-348 in place of the differential side bearings and a Dial Indicator C-3339. Before proceeding with the differential bearing preload and gear backlash measurements, measure the pinion gear depth and prepare the pinion for installation. Establishing proper pinion gear depth is essential to establishing gear backlash and tooth contact patterns. After the overall shim thickness to take up differential side play is measured, the pinion is installed, and the gear backlash shim thickness is measured. The overall shim thickness is the total of the dial indicator reading and the preload specification added together. The gear backlash measurement determines the thickness of the shim used on the ring gear side of the differential case. Subtract the gear backlash shim thickness from the total overall shim thickness and select that amount for the pinion gear side of the differential (Fig. 14). Differential shim measurements are performed with spreader W-129-B removed.

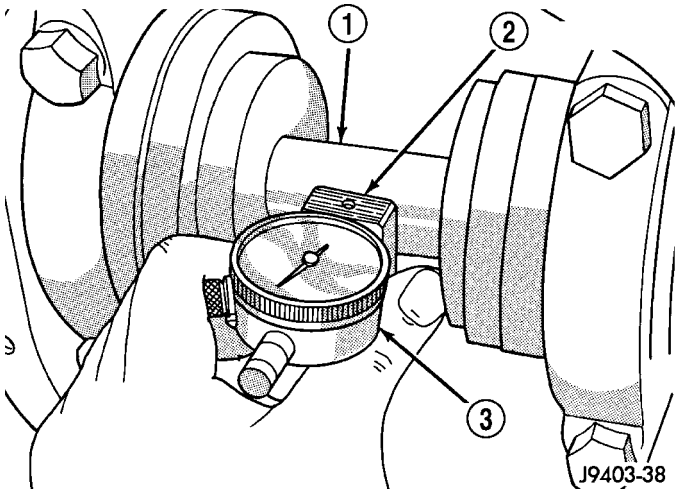



*Fig. 13 PINION DEPTH MEASUREMENT*

1 - ARBOR
2 - SCOOTER BLOCK
3 - DIAL INDICATOR

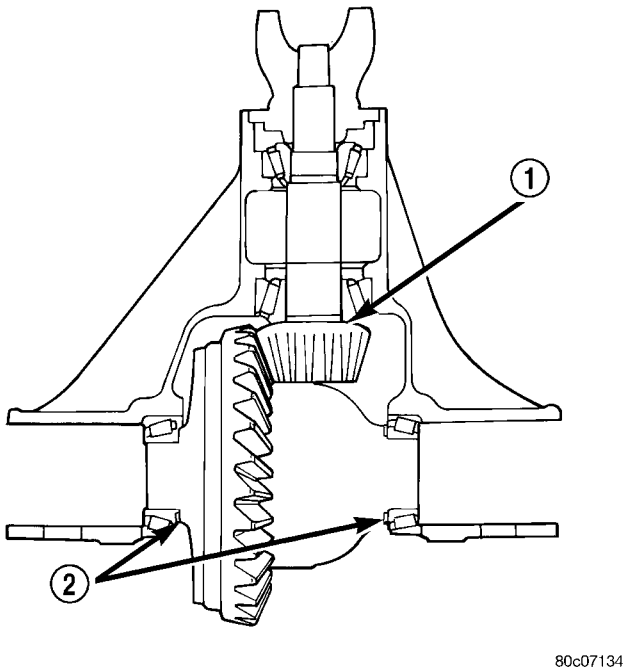



*Fig. 14 SHIM LOCATIONS*

1 - PINION GEAR DEPTH SHIM/OIL SLINGER
2 - DIFFERENTIAL BEARING SHIM